PEO Ottawa Chapter

Robotics Competition

RULES AND GUIDELINES

Unmanned Ground Vehicle Design Challenge

1. Purpose and Educational Intent

The PEO Ottawa Chapter Robotics Competition is intended for college and university students in the Ottawa region. Its purpose is to increase awareness of Professional Engineers Ontario (PEO) and the engineering profession while giving students a practical opportunity to apply engineering knowledge in a multidisciplinary design-and-build project.

The competition is not intended to be only a race. Teams should demonstrate how they define engineering problems, make design decisions, build and integrate a working system, test it, control cost, communicate results, and consider whether the design could be produced at a larger scale.

The challenge should encourage learning in areas such as mechanical design, electronics, programming, control systems, sensing, robotics, testing, cost management, technical communication, and engineering judgement.

1.1 Guiding principles

- Engineering first: evaluation should reward sound engineering decisions, not paperwork volume.
- Safe and ethical operation is mandatory and should be treated as a baseline requirement rather than a source of bonus points.
- Fairness: all teams should work under the same published technical and cost constraints.
- Low administrative burden: teams should provide only the information judges and organizers actually need.
- Transparent judging: criteria should be short, understandable, and published before the event.
- Learning over perfection: the event should reward testing, iteration, and thoughtful trade-offs, including low-cost solutions.

2. Competition Challenge

Each team will design and build a physical unmanned ground vehicle (UGV) capable of transporting a 1 kg payload through a predetermined obstacle course.

The vehicle is expected to demonstrate both remote-controlled and autonomous operation. The final event format should make it possible to observe and evaluate both modes without requiring separate administrative submissions.

2.1 Core vehicle functions

- Transport a 1 kg payload from the start area to the designated destination.
- Operate by remote control when required by the event format.
- Operate autonomously when required by the event format, in accordance with the definition of "Autonomous Mode" provided in Appendix C.
- Carry an onboard camera that provides a real-time forward-facing view in the vehicle's direction of travel during operation. The live feed must be viewable by the team and may also be used by the autonomous navigation system.
- Remain stable enough to complete the course without dropping or damaging the payload.
- Be presented as a working physical prototype.

2.2 Course concept

The course may include obstacles, turns, inclines, narrow sections, and other features selected by the organizing committee. Performance should consider successful completion, reliability, navigation, payload integrity, and the degree of human intervention, not only elapsed time.

Committee decision required

Final course dimensions, obstacle types, start/finish procedure, number of attempts, time limits, and the exact distinction between remote-controlled and autonomous runs remain to be defined.

3. Participants and Teams

The intended participants are students enrolled in colleges and universities in the Ottawa region. Final eligibility requirements, team size, faculty/advisor involvement, and any institution-specific limits remain to be approved by the organizing committee.

The rules should avoid unnecessary eligibility documentation. A simple team registration identifying the institution, team members, and one primary contact should normally be sufficient unless an additional requirement is needed for the venue or event administration.

4. Technical Requirements

Requirement	Working draft rule
Vehicle type	Physical unmanned ground vehicle (UGV) prototype.
Payload	Must transport the organizer-supplied 1 kg open-top competition payload as defined in Appendix C.

Requirement	Working draft rule
Control	Must support remote-controlled operation and autonomous operation in accordance with Appendix C.
Camera	Must provide a real-time forward-facing onboard view during operation. The live feed must be viewable by the team and may also be used by the autonomous navigation system.
Safety	Must be safe and ethical to operate in the competition environment.
Vehicle envelope	Maximum external envelope volume: 0.040 m^3 (40 L), calculated as Length × Width × Height. No single external dimension may exceed 700 mm.
Vehicle mass	Maximum 15 kg, excluding the 1 kg competition payload.
Other technical limits	Power, energy source, communications, sensors, and any prohibited systems are TBD unless later specified by the committee.
Commercial components	Commercially available RC vehicles, chassis, modules, and components may be used or modified, provided applicable costs are included in the BOM and the completed vehicle complies with all competition requirements.

4.1 Vehicle size and mass limits

The vehicle size and mass limits are intended to keep the challenge proportional to the 1 kg payload and to reward compact, efficient engineering rather than solutions that rely primarily on vehicle scale.

- Maximum external envelope volume: 0.040 m^3 (40 L), calculated as V = Length × Width × Height.
- No single external dimension may exceed 700 mm.
- The envelope is the smallest rectangular box that contains the vehicle in its competition configuration. If the vehicle changes geometry during a run, it must remain within the volume and single-dimension limits throughout the run.
- The 1 kg competition payload itself is excluded from the envelope calculation. Payload supports, containers, brackets, and other hardware that form part of the vehicle are included.
- Maximum vehicle mass: 15 kg, excluding the 1 kg competition payload. The measured vehicle mass includes batteries, structure, electronics, sensors, camera, payload-support hardware, and all other onboard components used during the run.
- No separate density limit applies. The combination of the 15 kg mass limit and 0.040 m^3 envelope limit corresponds to 375 kg/m^3 only when both maximum values are reached; this value is informative and is not an additional design constraint.

To minimize administration, envelope and mass compliance should normally be checked once during the pre-run technical inspection. Rechecking is only necessary if the vehicle is materially modified or if compliance is reasonably in question.

4.2 Recommended minimum safety provisions

To keep the safety process simple, the event should use a short pre-run technical check rather than a large safety submission. The following provisions are recommended for committee approval:

- The vehicle must have a readily accessible method to stop or disable motion.
- The payload container must be securely supported by the vehicle. However, the open top of the organizer-supplied payload container must remain unobstructed, and no lid, cover, retaining device, or other mechanism may be used to prevent loss of its contents.
- The vehicle must not present obvious hazards from loose wiring, unstable components, sharp projections, overheating, or uncontrolled motion.
- Organizers or judges may stop a run immediately if continued operation appears unsafe.
- A safety stop should not create a paperwork requirement; the incident should only be noted if it affects scoring, eligibility to rerun, or future operation.

4.2.1 Battery Safety Requirements

All teams must follow basic battery safety practices during the competition. Particular attention must be given to lithium-based batteries, which can present fire and thermal risks if damaged, improperly charged, or incorrectly handled.

The following requirements apply:

- Batteries must be charged using a charger specifically designed and approved for the battery type and chemistry being used.
- Batteries must be charged and stored in a safe location, away from flammable materials.
- Damaged, swollen, leaking, overheated, or otherwise compromised batteries must not be used.
- Each vehicle must have a clearly accessible **physical power disconnect or master ON/OFF switch** capable of immediately removing electrical power from the propulsion system.
- A software command, software shutdown function, or remote-control command alone does **not** satisfy the requirement for a physical power disconnect.
- Teams must ensure that the vehicle is powered off whenever it is being inspected, serviced, transported, or handled outside the designated operating area.
- Competition officials may prohibit the use of any battery or electrical system considered unsafe.

5. Budget and Cost Control

The competition should encourage clever, cost-effective engineering. A target build cost of CAD 100 is proposed as an aspirational objective, while the hard maximum build cost is CAD 500 including applicable taxes.

Cost rule	Draft interpretation
Target budget	CAD 100. Teams are encouraged, but not required, to meet this target.
Hard limit	CAD 500 including applicable taxes.
Purchased parts	Count at actual cost including applicable taxes.

Cost rule	Draft interpretation
Donated/sponsored parts	Count at a reasonable market value.
Reused parts	Count at a reasonable market value to preserve fairness among teams.
Cost record	Submit a concise bill of materials (BOM) and total project cost.
Supporting evidence	Teams should retain receipts or equivalent evidence. Routine submission of every receipt is not necessary unless the organizers request verification.

6. Team Deliverables

Deliverables should give judges enough evidence to understand the engineering work without requiring a formal report package. The following set combines the items already proposed during committee discussions and keeps each requirement practical.

Deliverable	Purpose / minimum expectation
Working prototype	The UGV used in the competition demonstration.
Design record	A concise design notebook or equivalent record showing major design decisions, alternatives considered, and key calculations or reasoning.
Bill of materials	Parts/components, quantities, estimated or actual values, and total build cost.
Testing results	Evidence of meaningful testing and what the team learned or changed as a result.
Poster	A clear summary of the problem, design, system architecture, testing, cost, and results.
5-minute presentation	A short oral explanation of the project and key engineering decisions.
Demonstration	Operation of the vehicle in the competition course or designated demonstration area.
Manufacturing feasibility note	A brief explanation of how the design could be produced at approximately 1,000 units per year, including likely production issues and cost considerations. This may be incorporated into the design record or poster rather than submitted as a separate report.

Format should not be judged for its own sake. Teams may organize the design record in the way that best explains their work, provided the required information is easy for judges to find.

7. Event Flow

A simple event sequence is recommended so that teams and judges spend most of their time on engineering and demonstration rather than administration.

1. Team check-in and confirmation of registration.
2. Brief technical/safety inspection of the vehicle.
3. Poster/design review and 5-minute team presentation.

4. Vehicle demonstration and competition run(s).
5. Judge scoring and, if necessary, a short clarification question to the team.
6. Awards and results announcement.

The inspection should verify only critical compliance items such as the physical vehicle, payload capability, vehicle envelope and mass limits, obvious safety concerns, and cost declaration. It should not become a second judging process.

8. Judging Framework

The preliminary proposal grouped evaluation into Concept and Design, Performance, and Technical and Manufacturing Feasibility. Committee discussion also proposed judging creativity, functionality, testing, engineering process, cost effectiveness, and presentation. The following 100-point framework is recommended as a simple consolidation of those ideas.

Criterion	Points	What judges should look for
Engineering concept & design rationale	20	Quality of the concept; integration of mechanical, electrical, software/control elements; appropriateness of design decisions; evidence of engineering judgement.
Creativity & innovation	15	Originality, resourcefulness, and whether the team found an effective solution rather than simply maximizing spending.
Functional performance	25	Course completion, payload integrity, stability, navigation, reliability, and demonstrated capability in the required operating modes.
Testing & validation	15	Quality of testing, use of results, iteration, and evidence that the team understands performance limitations.
Cost effectiveness & manufacturing feasibility	15	Credibility of the BOM, intelligent cost choices, and practical consideration of production at about 1,000 units per year.
Communication	10	Clarity of the poster, 5-minute presentation, demonstration explanation, and ability to answer concise judge questions.

8.1 Judging approach

- Judges should score the engineering work, not the volume or visual polish of paperwork.
- Safety and compliance with hard rules should be treated as eligibility/pass-fail matters, not as optional bonus points.
- A low-cost solution may score better than a more expensive solution if it demonstrates stronger engineering judgement and comparable or better performance.
- Judges may ask short clarification questions when needed, but teams should not be required to produce additional documents during judging.
- The final scoring method for timed course performance, payload loss, interventions, penalties, reruns, and ties must be published before the event. Course completion time must not override successful completion, payload integrity, stability, navigation, and reliability as the principal functional-performance considerations.

9. Awards

One overall winner may be selected from the final approved scoring system. The committee may also use special awards to recognize different forms of engineering achievement without forcing all teams into a single definition of success.

Awards suggested for committee consideration include:

- Best Innovation
- Best Under-$100 Solution
- Best Social Impact
- Best Sustainability Solution
- People's Choice Award
- Most Likely to Become a Startup

The committee does not need to offer every special award. A smaller set may be selected based on the number of teams, available sponsorship, and event objectives.

10. Event Conduct and Judge Authority

Teams are expected to act safely, ethically, and respectfully toward other participants, judges, volunteers, and venue staff. Judges and organizers should apply the published rules consistently and use technical judgement where a situation is not explicitly covered.

To avoid unnecessary process, routine judging decisions should be resolved at the event. If a team believes a material scoring or rule interpretation error occurred, the team lead may raise the issue promptly with the designated head judge or competition lead. The matter should be reviewed using the published rules and available evidence, without a formal written appeal unless the organizing committee later determines that one is necessary.

11. Rules Clarifications and Changes

Before the competition, the organizing committee may issue clarifications needed to resolve ambiguity or finalize technical details. Any change that could materially affect a team design, budget, or competition strategy should be communicated to all registered teams through the same official channel.

The goal is to maintain one clear source of truth for participants. Organizers should avoid parallel versions of the rules or informal requirements that are known only to some teams.

12. Committee Decisions Still Required Before Final Publication

Open item	Decision needed
Participant eligibility	Eligible institutions, student status, team size, and advisor role.
Power and communications	Any limits on batteries, voltage, wireless systems, external computing, or off-board control.

Open item	Decision needed
Course	Dimensions, surface, obstacle types, start/finish zones, payload placement, and whether teams may inspect or practice on the course.
Run format	Number of runs/attempts, remote-controlled vs autonomous sequence, time limits, rerun conditions, manual-recovery treatment, and technical-failure treatment.
Performance scoring	Objective formula for course completion, time, payload loss, human intervention, penalties, reruns, and ties; confirm whether completion time will be used as a tie-breaker.
Technical inspection	Final one-page checklist and any non-negotiable safety requirements; inspection should include the 40 L envelope and 15 kg mass checks.
Cost boundary	Whether general-purpose tools, laptops, test equipment, chargers, and similar non-vehicle items are excluded from the CAD 500 cap.
Judging panel	Number of judges, lead judge role, and how scores are combined.
Awards	Which special awards will actually be offered and any award-specific eligibility rules.
Event logistics	Venue, indoor/outdoor environment, schedule, registration deadline, number of teams, sponsorship, and prize budget.
Payload standardization	Confirm the final loose material and any acceptable mass tolerance for the nominal 1.00 kg competition payload.
Judging rubric	Develop scoring anchors/descriptors for each judging category to improve consistency while preserving reasonable judge discretion.
Insurance and liability	Confirm PEO insurance coverage, venue requirements, participant waivers, and any required consent or release forms.

Appendix A - One-Page Team Readiness Checklist

This checklist is intentionally short. It may be used by teams before arrival and by organizers during check-in.

☐ Team registration completed.

☐ Physical UGV prototype is present and operational.

☐ Vehicle can carry the organizer-supplied 1 kg open-top payload without obstructing its open top.

☐ Vehicle external envelope is no more than 0.040 m^3 (40 L) and no single external dimension exceeds 700 mm.

☐ Vehicle mass is no more than 15 kg, excluding the 1 kg competition payload.

☐ Remote-controlled mode is ready for demonstration.

☐ Autonomous mode is ready for demonstration.

☐ Onboard camera provides the required real-time forward-facing view during operation and the team can view the live feed.

☐ Vehicle has a practical means to stop/disable motion.

☐ No obvious safety hazards are present.

☐ BOM and total project cost are available; declared cost does not exceed CAD 500 including taxes.

☐ Design record is available.

☐ Testing results are available.

☐ Poster is ready.

☐ 5-minute presentation is ready.

☐ Manufacturing feasibility at approximately 1,000 units/year is addressed in the design record or poster.

Appendix B - Proposed Judge Score Sheet

Judges can enter a single point value for each category. The short comment field is optional and should be used mainly when it helps explain a high/low score or supports feedback to the team.

Criterion	Max	Score	Optional short note
Engineering concept & design rationale	20		
Creativity & innovation	15		
Functional performance	25		
Testing & validation	15		
Cost effectiveness & manufacturing feasibility	15		

Criterion	Max	Score	Optional short note
Communication	10		
TOTAL	100		

Safety/compliance status: PASS / HOLD FOR REVIEW

Team number/name: _____

Judge: _____ Date: _____

Appendix C – Working Definitions

Term	Working definition
UGV	Physical Unmanned Ground Vehicle developed by the team for the challenge.
Remote-controlled mode	Vehicle actively controlled by a human operator using the permitted control system.
Autonomous mode	**Autonomous Mode** In **Autonomous Mode**, the vehicle must be capable of navigating a predetermined course **without any human intervention or external control** from the moment the autonomous run begins until the vehicle completes the course. The only human interventions permitted during the autonomous run are: • **Emergency Stop:** The operator may immediately stop the vehicle if there is a safety concern, risk of collision, equipment malfunction, or any other situation requiring an immediate halt. • **Manual Reset/Recovery:** If authorized by the judges, the vehicle may be manually repositioned or reset after becoming immobilized or leaving the designated course. Any such intervention may result in a time or scoring penalty, as defined in the final competition rules.

Term	Working definition
Payload	The 1 kg load the vehicle must transport through the course. **Competition Payload** The competition payload will consist of an **open-top rigid container**, supplied by the organizers, carrying a standardized loose material. The container will be approximately **100 mm × 100 mm × 100 mm** and may be manufactured from ABS using 3D printing. The payload will have a total nominal mass of **1.00 kg**, measured immediately before the start of each competition run. The container will remain **open during the entire run**, and teams may not install a lid, cover, retaining device, or other mechanism intended to prevent the contents from leaving the container. The competition payload will be manually placed on the vehicle by the team or competition officials before each run. The vehicle is not required to autonomously load or unload the payload. The loose payload material will be standardized by the organizers and will be the same for all teams. Dry sand is proposed as the standard material. The complete payload will be weighed: 1. **Immediately before the run; and** 2. **Immediately after the run.** The difference between the initial and final mass will be considered the **payload loss**. **Payload Loss = Initial Payload Mass - Final Payload Mass** Loss of material during the run may result in a performance penalty according to the final scoring system. The purpose of this requirement is to evaluate not only whether the vehicle completes the course, but also its **stability, control, smoothness of operation, and ability to transport a load without loss**.
BOM	Bill of Materials: parts/components, quantities, values, and total project cost.
Hard limit	A mandatory maximum. The working build-cost hard limit is CAD 500 including applicable taxes.
Target budget	An aspirational CAD 100 engineering target; it does not replace the hard limit.
Vehicle envelope	Smallest rectangular box containing the vehicle in competition configuration. Length × Width × Height must not exceed 0.040 m^3 (40 L), and no single external dimension may exceed 700 mm.
Vehicle mass	Operating vehicle mass excluding the 1 kg payload; includes batteries and all onboard components. Maximum: 15 kg.
Onboard camera	A forward-facing camera providing a real-time view in the vehicle's direction of travel during operation. The live feed must be viewable by the team and may also be used by the autonomous navigation system.
Commercial component	A commercially available RC vehicle, chassis, module, or component that may be used or modified by a team. Applicable cost must be included in the BOM, and the completed vehicle must comply with all competition requirements.

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